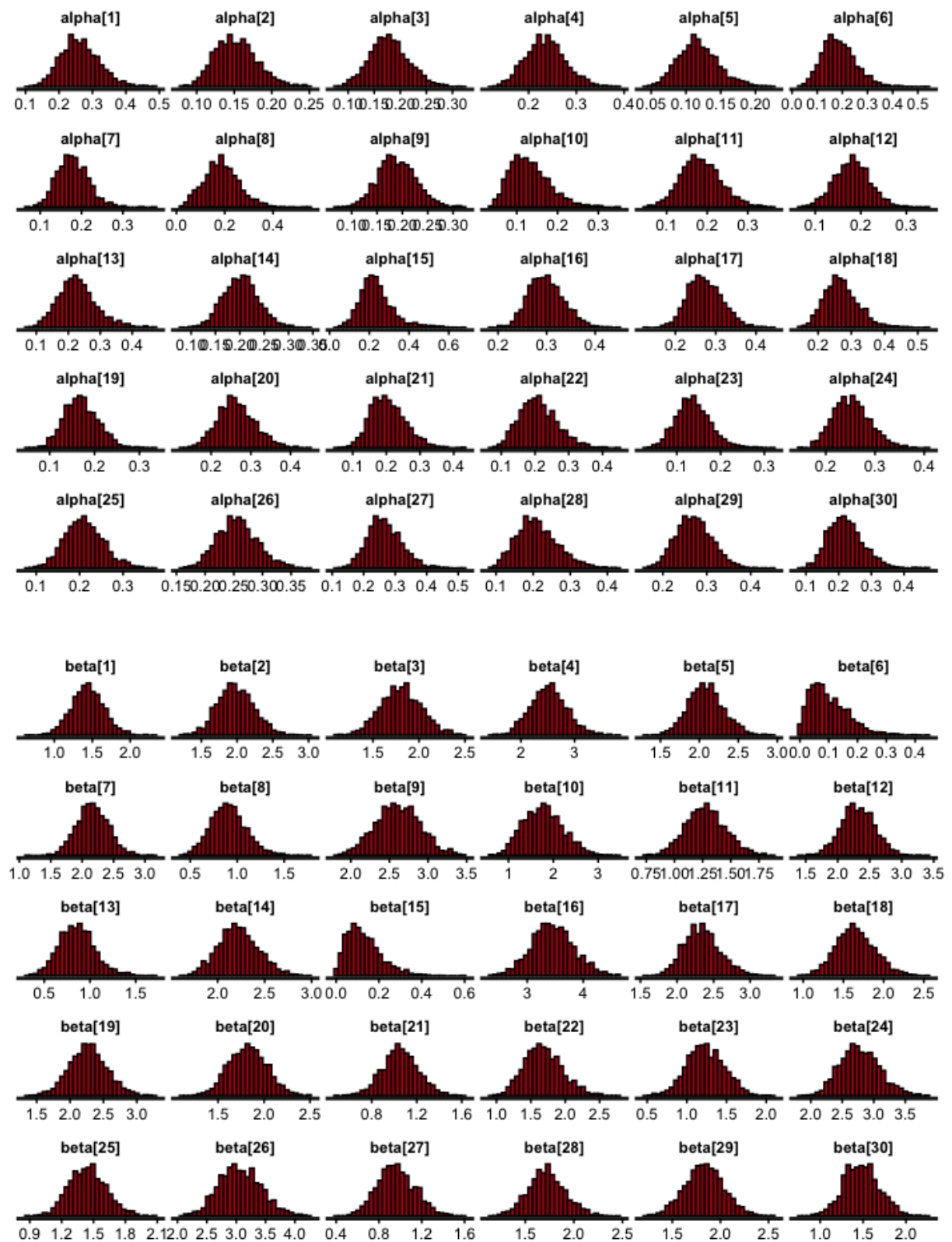
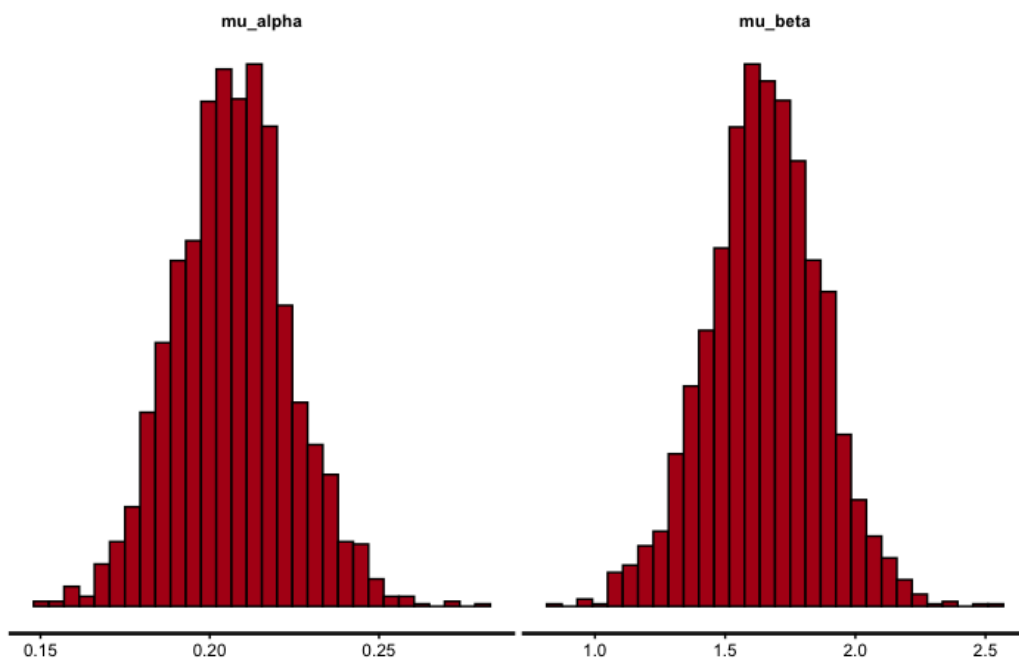


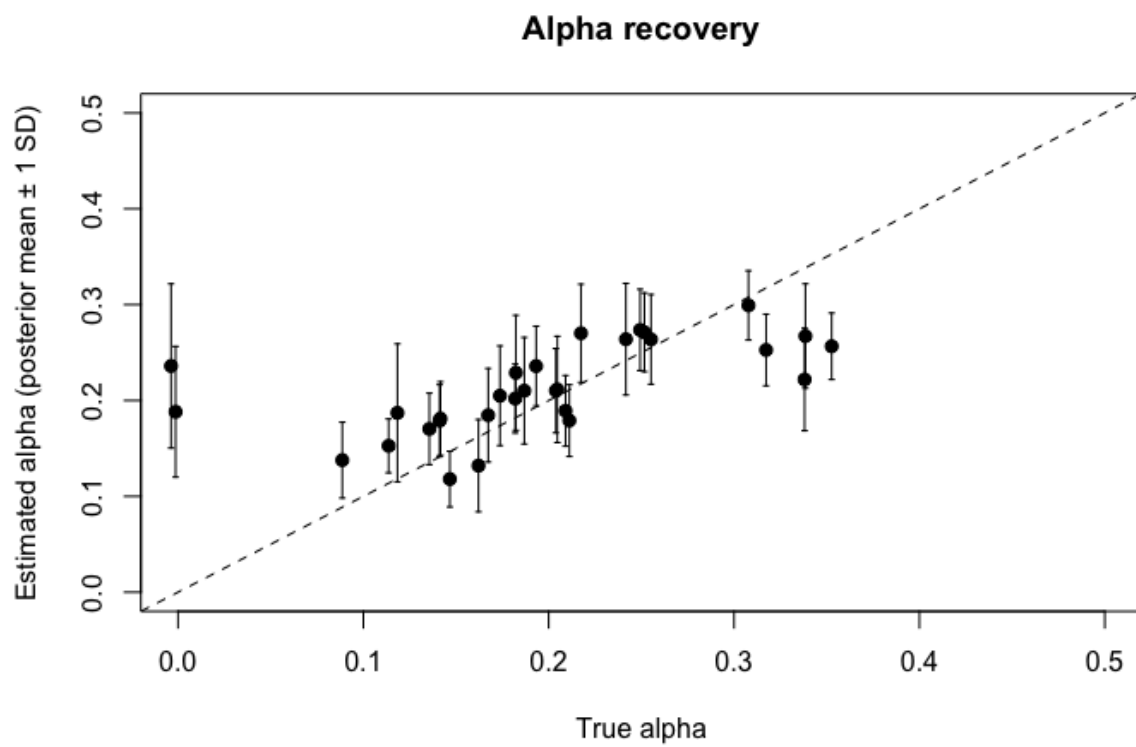
Q1 (2a) Posterior distribution of individual parameters alpha and beta

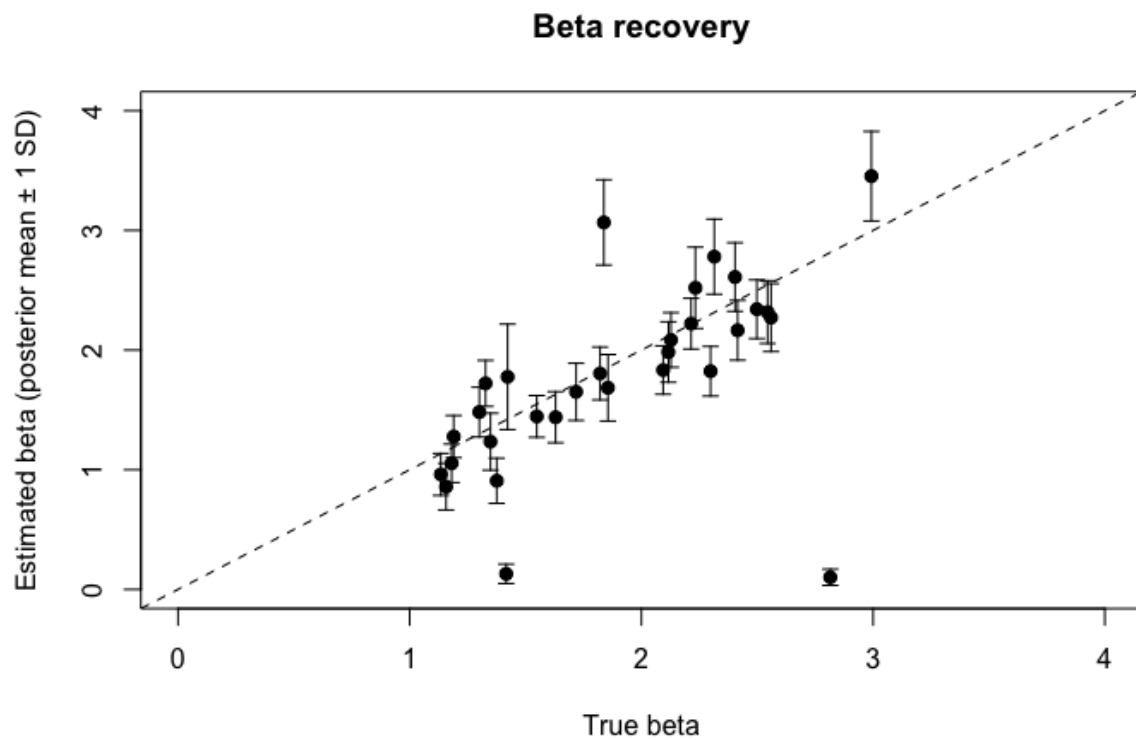


Posterior distribution of hyperparameters μ_{α} , μ_{β}



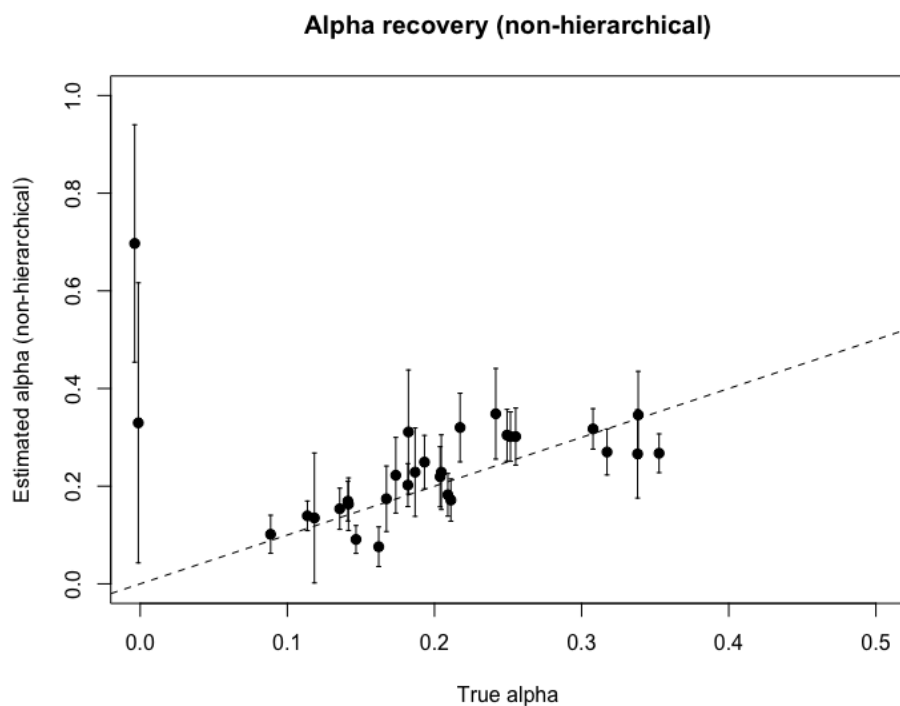
Q1 (2b) Scatterplot of true and estimated parameters

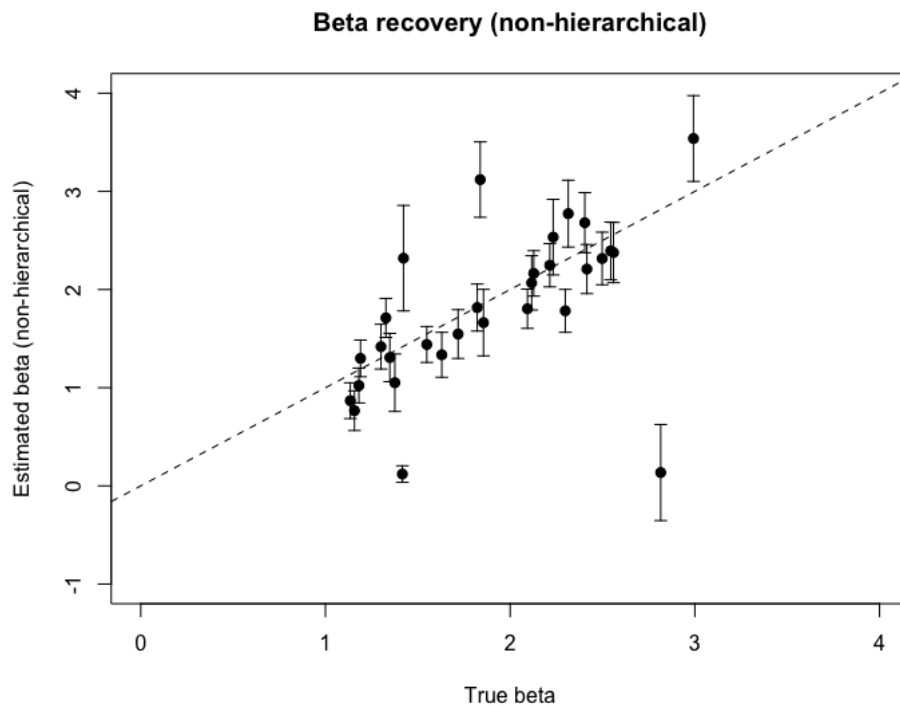




Looking at the plots, the true parameters seem relatively well-recovered as the posterior mean estimates cluster around the dashed identity line, and most error bars cross the diagonal as well. We can also see that there is mild to moderate shrinkage and a few subjects showing poor recovery.

Q2 (2) Scatter plot of true and estimated parameters using non-hierarchical model

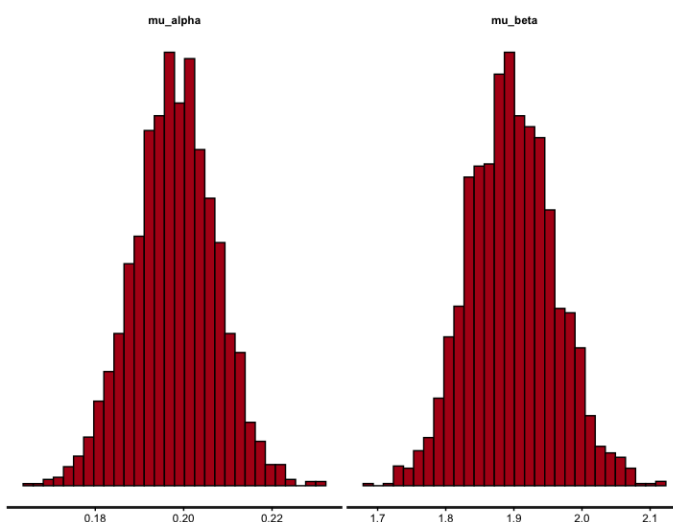




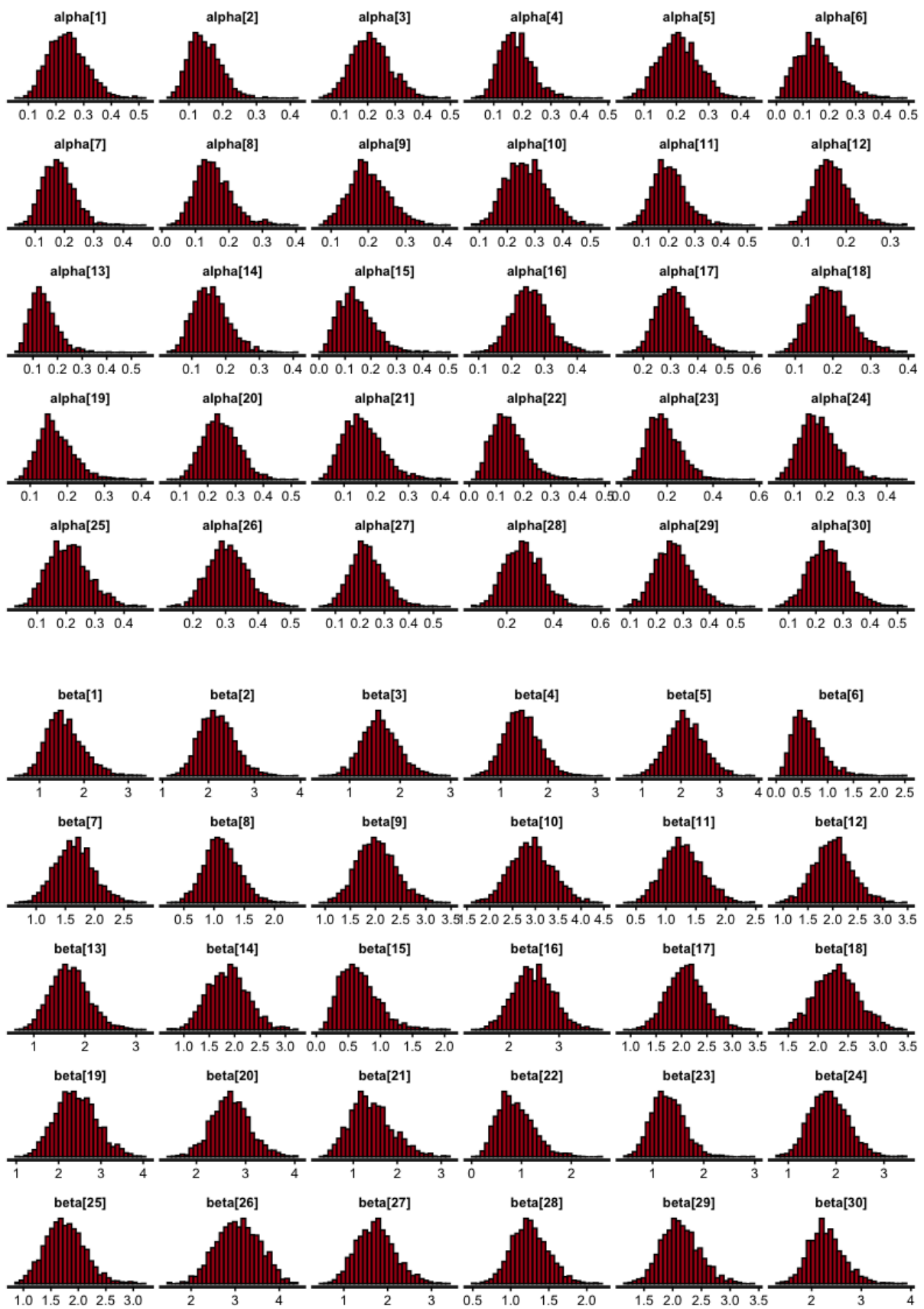
Q2 (3)

Comparing the hierarchical and non-hierarchical scatter plots, it seems that the data points are similarly clustered around the dashed identity line in the non-hierarchical plot as well. However, we can see that the error bar for one subject stretches to an extreme extent on the left side of the alpha recovery (non-hierarchical) figure. When a subject has a true parameter of $\alpha=0$, this implies that there is almost no value updating and the choices made by the subject contain no information about the parameter, leading to a flat likelihood. In a hierarchical model, these shortcomings are compensated by using partial pooling towards the group mean. However, in the non-hierarchical model, this does not exist, and may lead to a much larger posterior standard deviation.

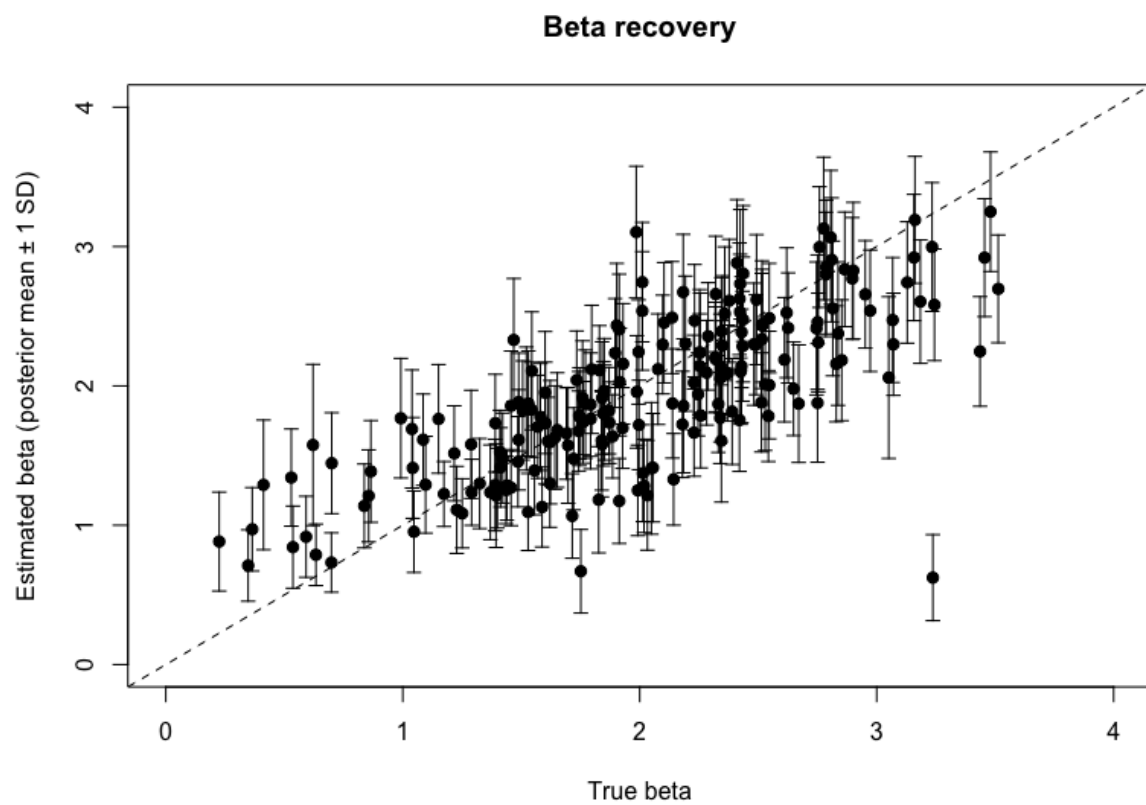
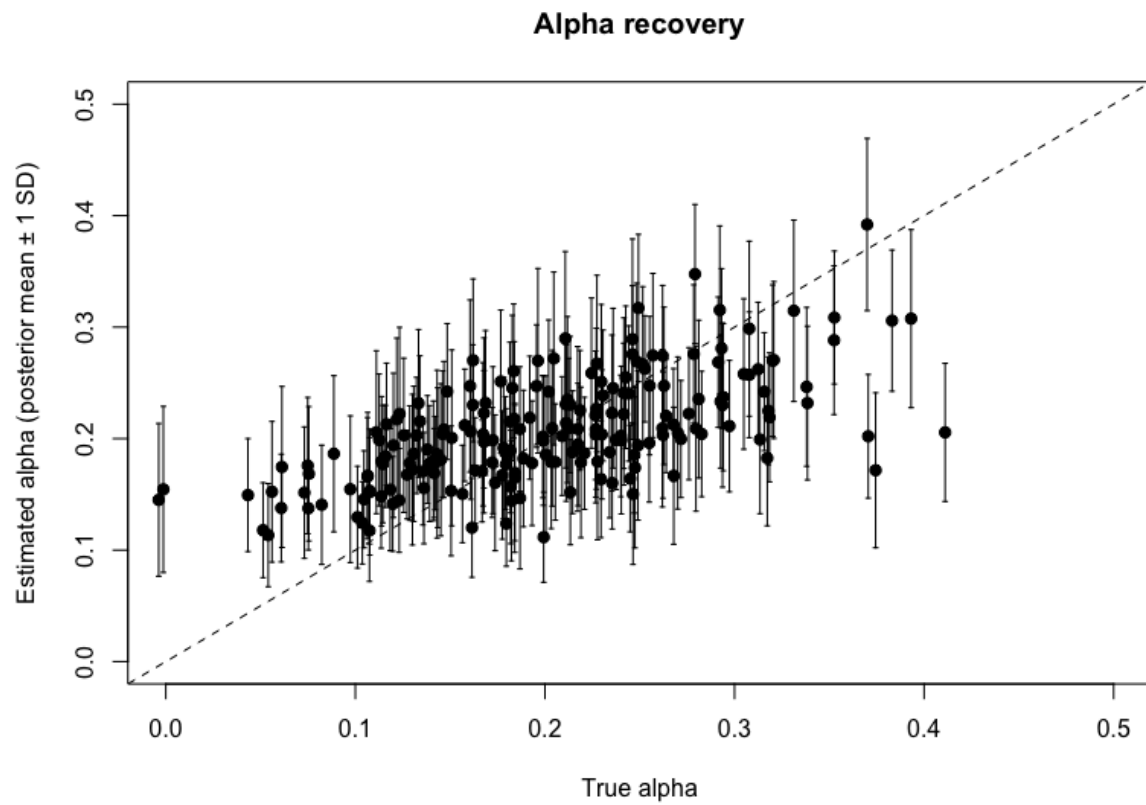
Q3 (2) Posterior distribution of hyperparameters μ_{α} , μ_{β} for 200 subjects



Posterior distribution of individual parameters for alpha and beta



Q3 (3) Scatterplot of true and estimated parameters for alpha and beta:

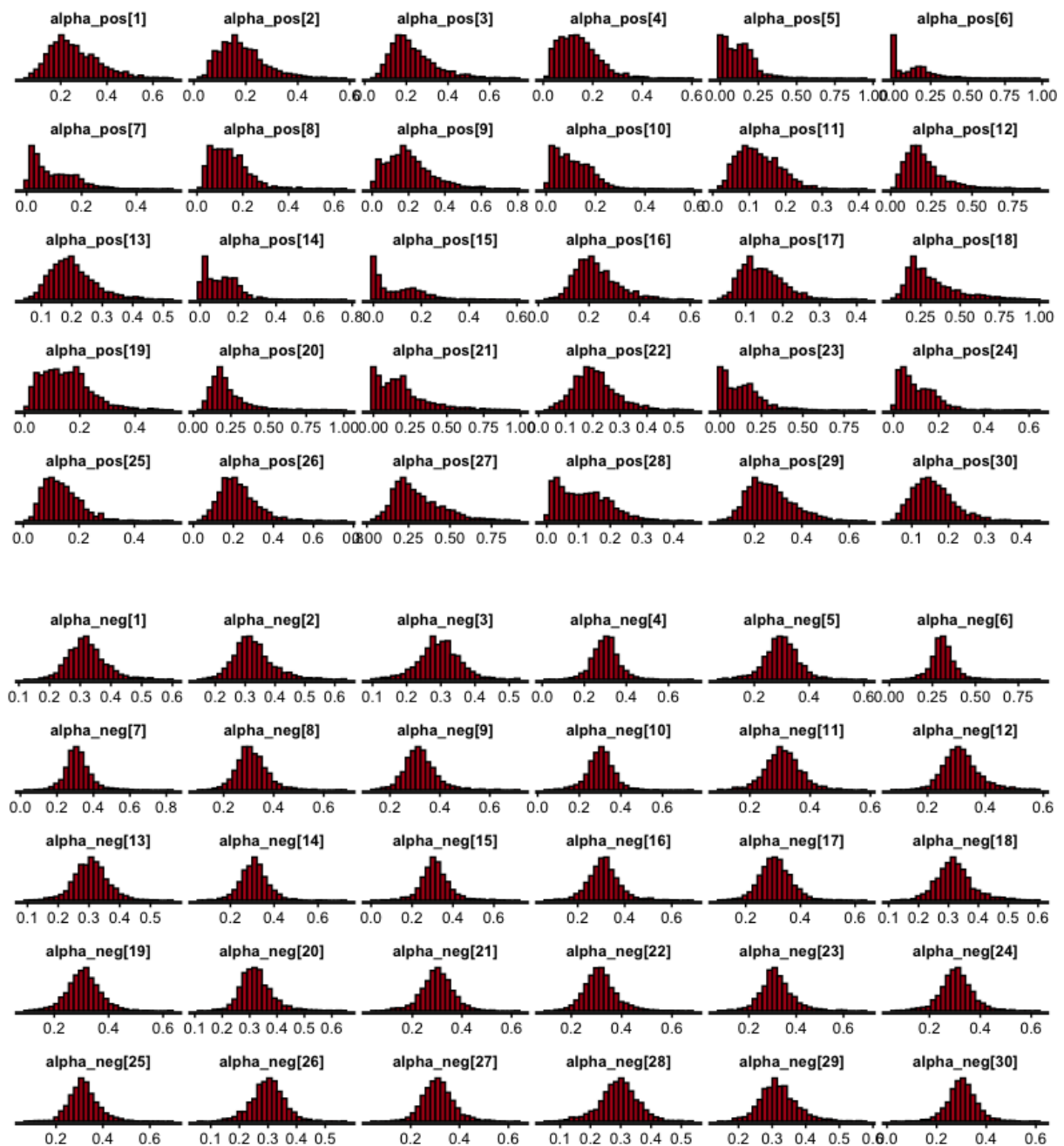


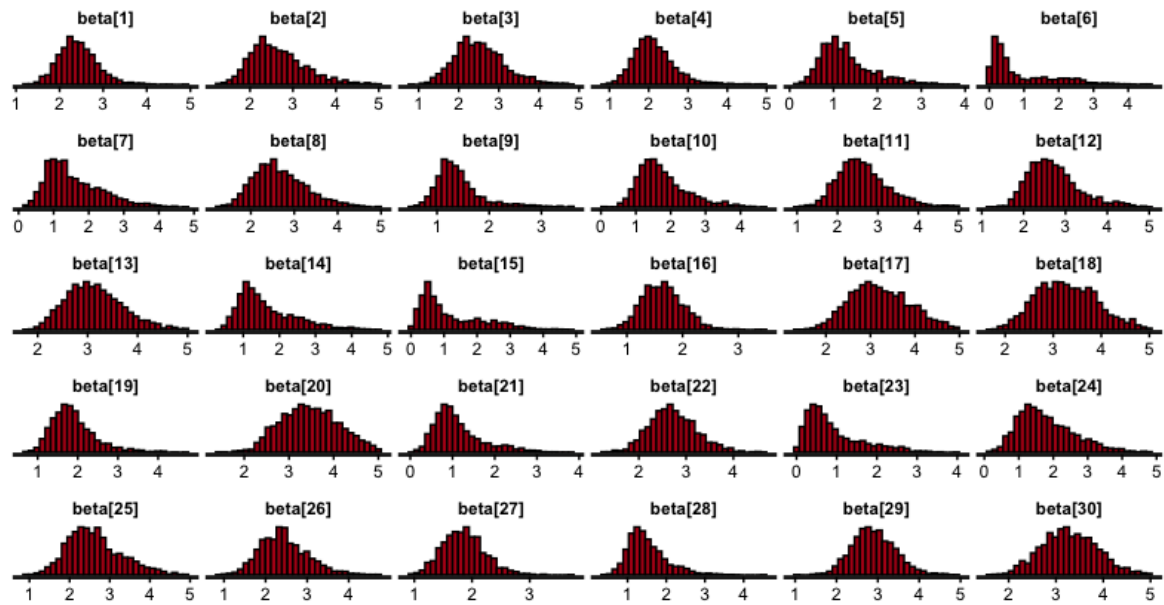
Q3 (4)

Looking at the plots, the true parameters seem relatively well-recovered as the posterior mean estimates cluster around the dashed identity line, and most error bars cross the diagonal as well.

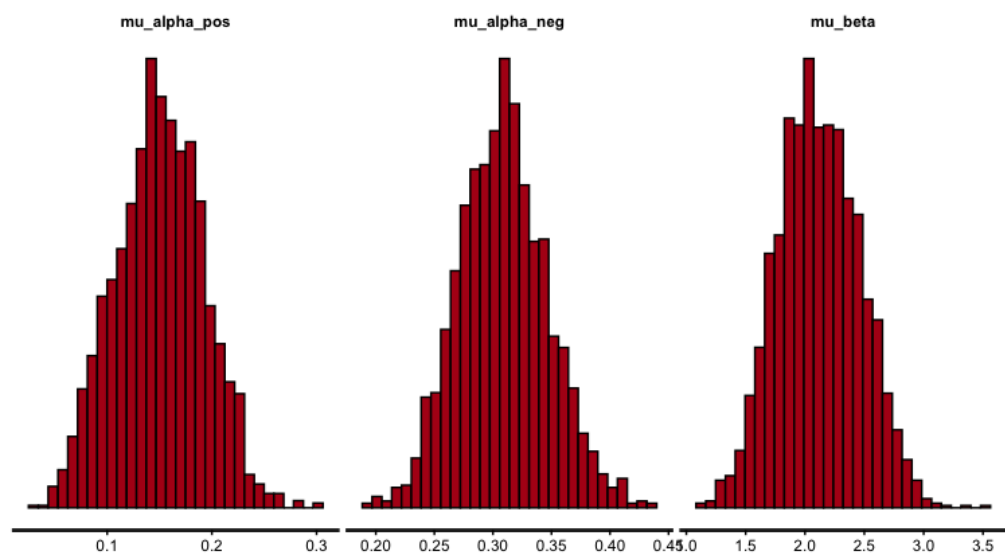
Between the two parameters, it seems that the alpha parameter contains more “shrinkage”. This may have occurred because the data constrains alpha less strongly compared to beta. In other words, the choice data may have less information about the learning rate alpha parameter than choice consistency (beta).

Q4 (1) Posterior distribution of individual parameters:

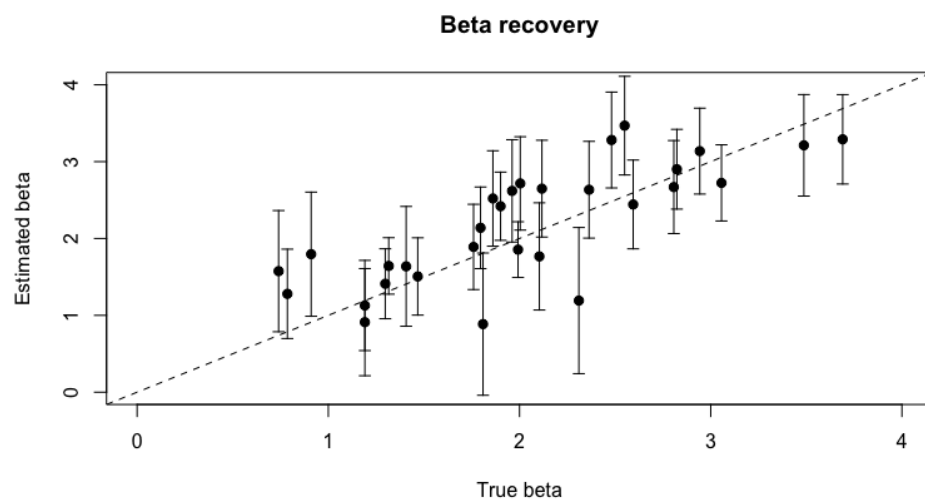
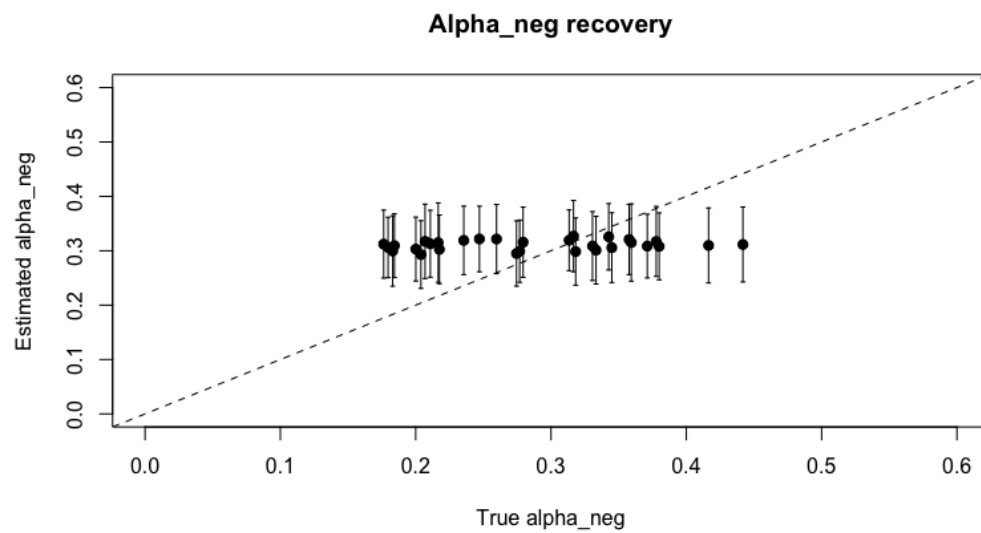
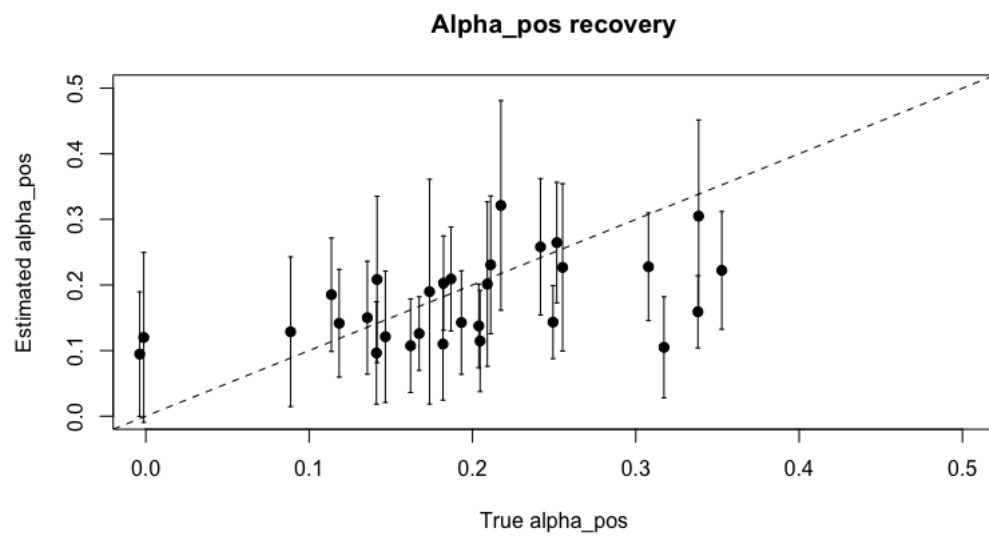




Posterior distribution of group parameters

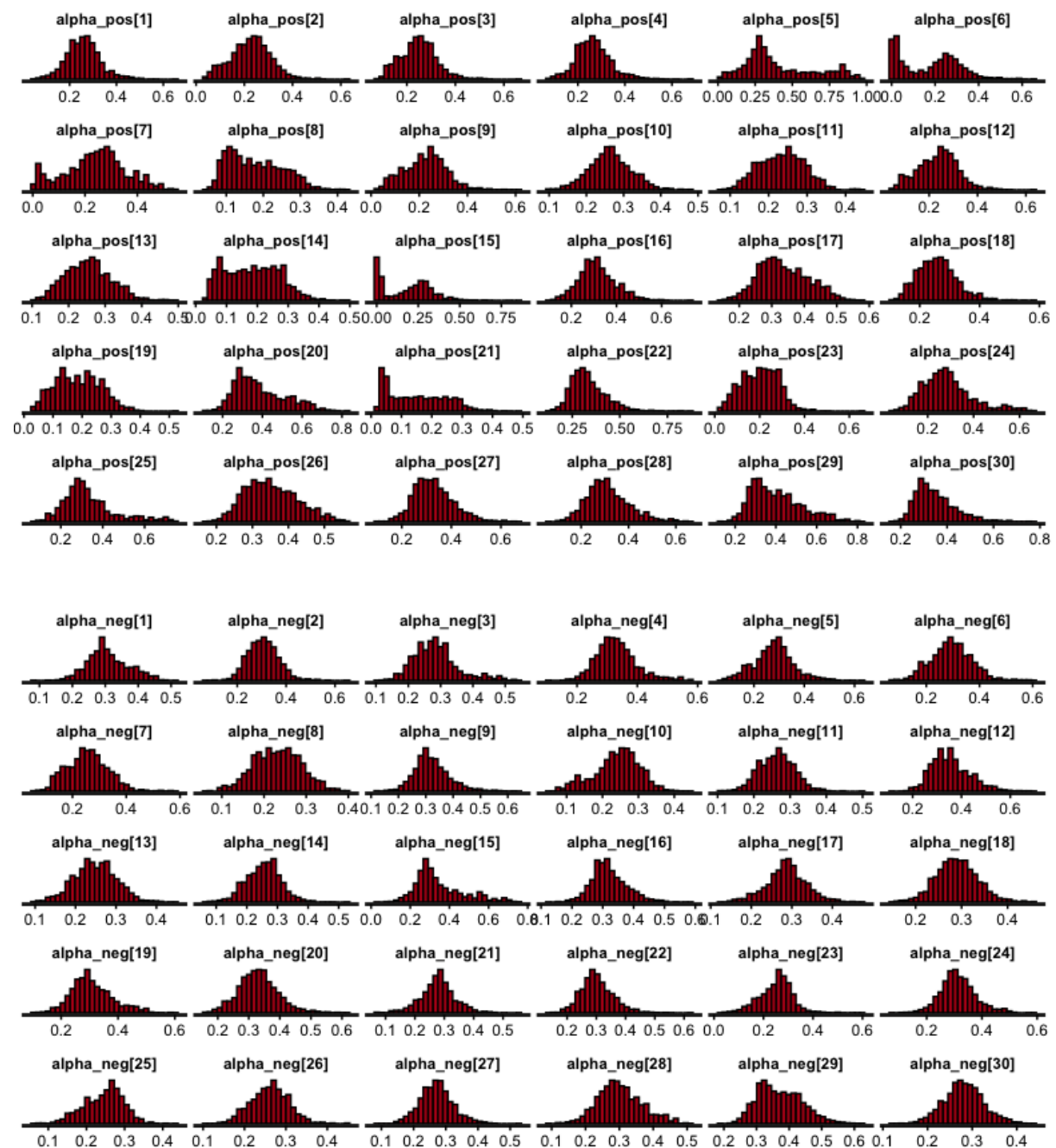


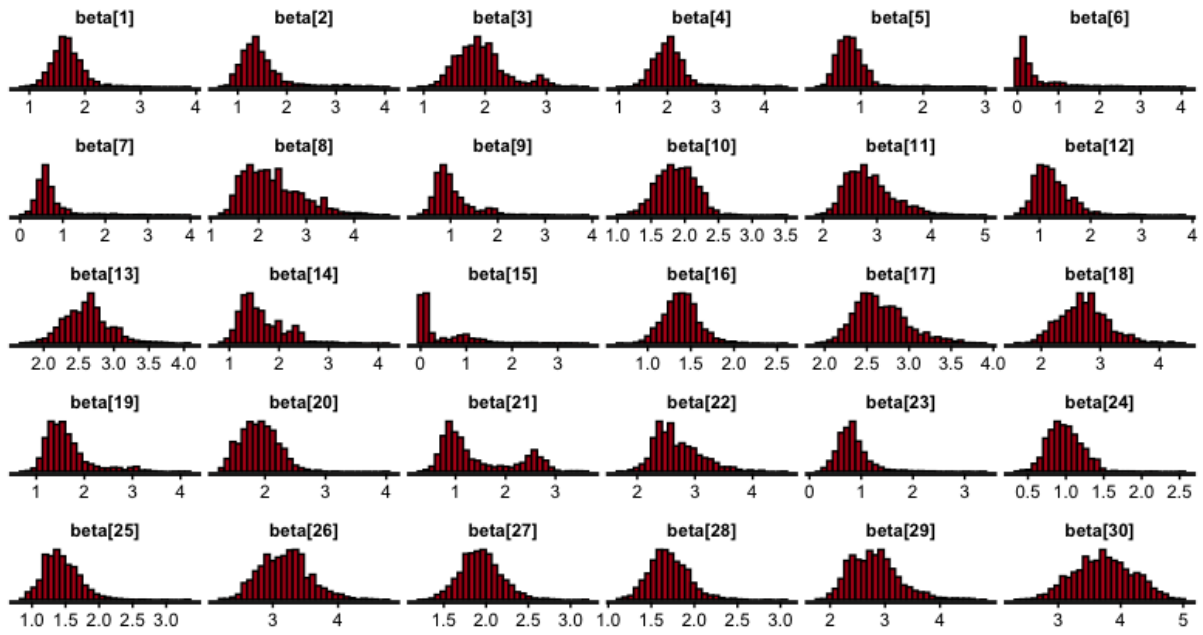
Q4 (2) Scatterplot of true and estimated parameters:



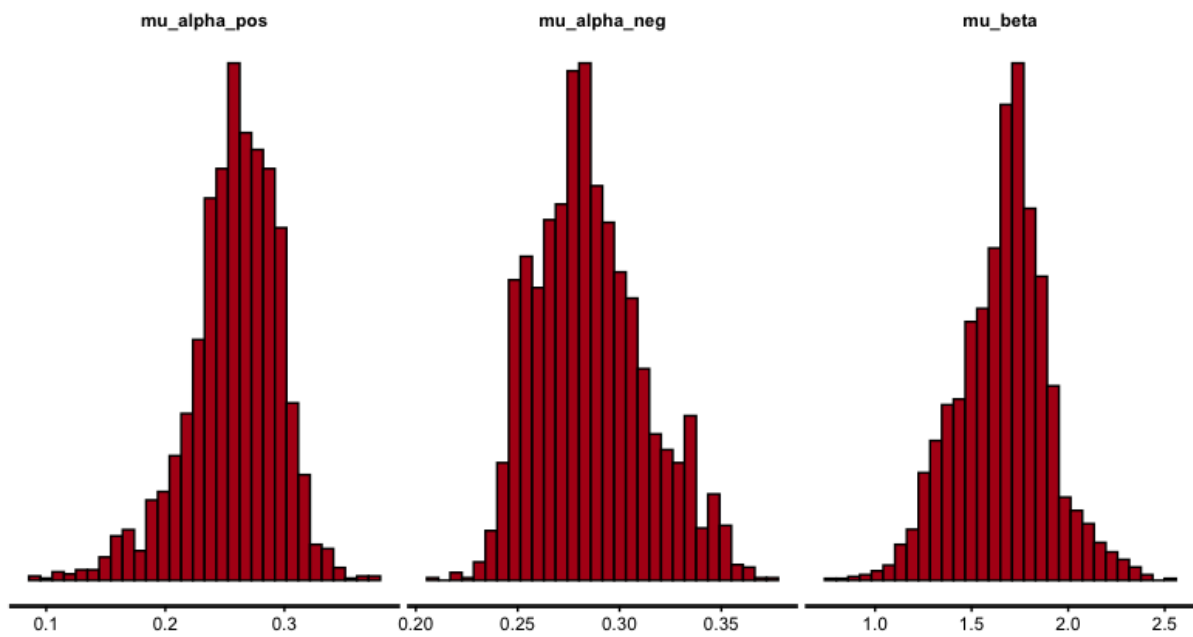
It seems that the beta parameter is relatively well-recovered. However, alpha_pos and alpha_neg show poor recovery. Alpha_neg especially shows extremely poor recovery, with most value estimates stuck at around approximately 0.30~0.32. This may have occurred because there are very few negative PE trials per subject, making it hard to estimate the parameter value using the data.

Q4 (3) Posterior distribution of individual parameters

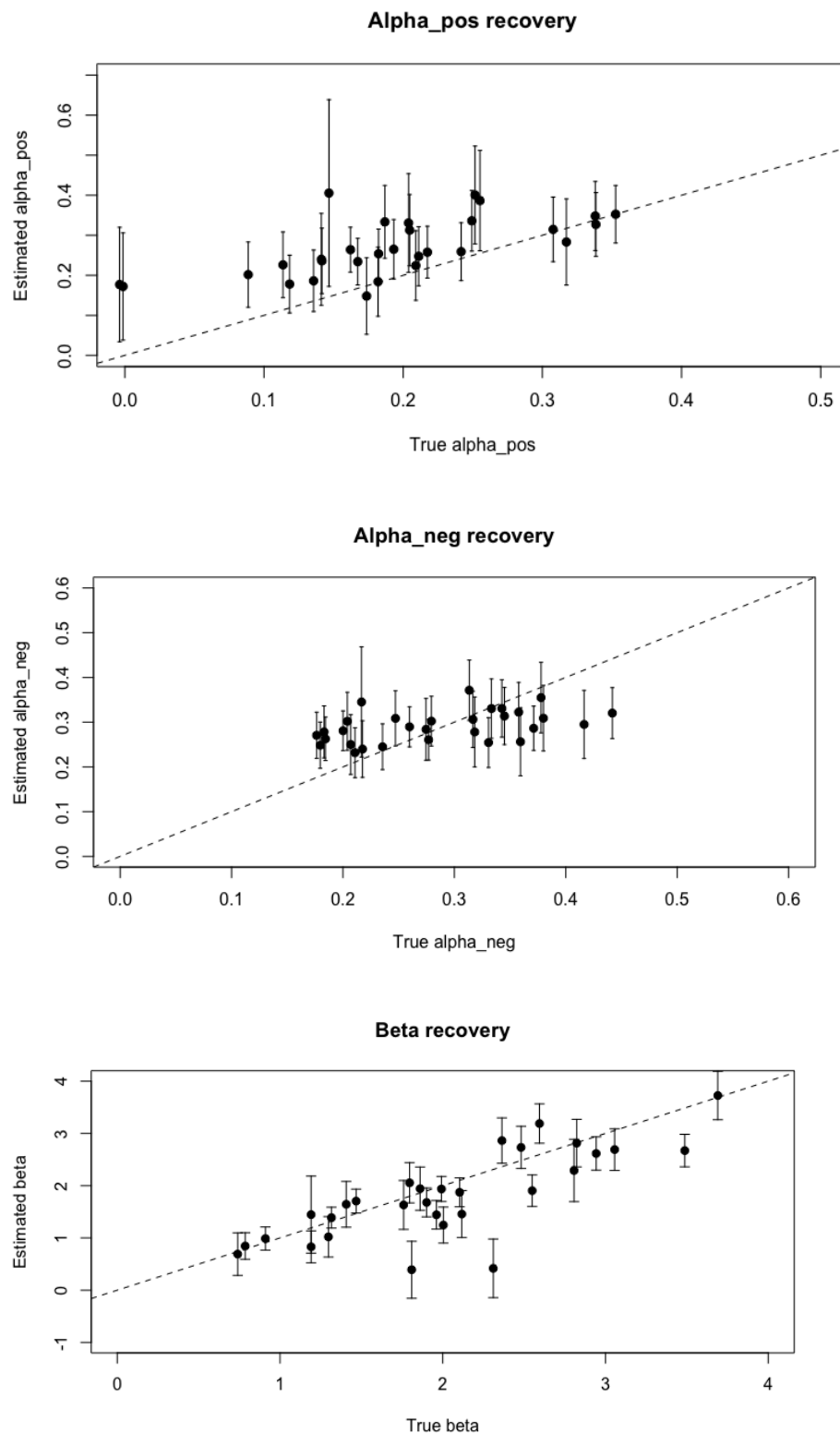




Posterior distribution of group parameters



Scatter plot of true and estimated parameters



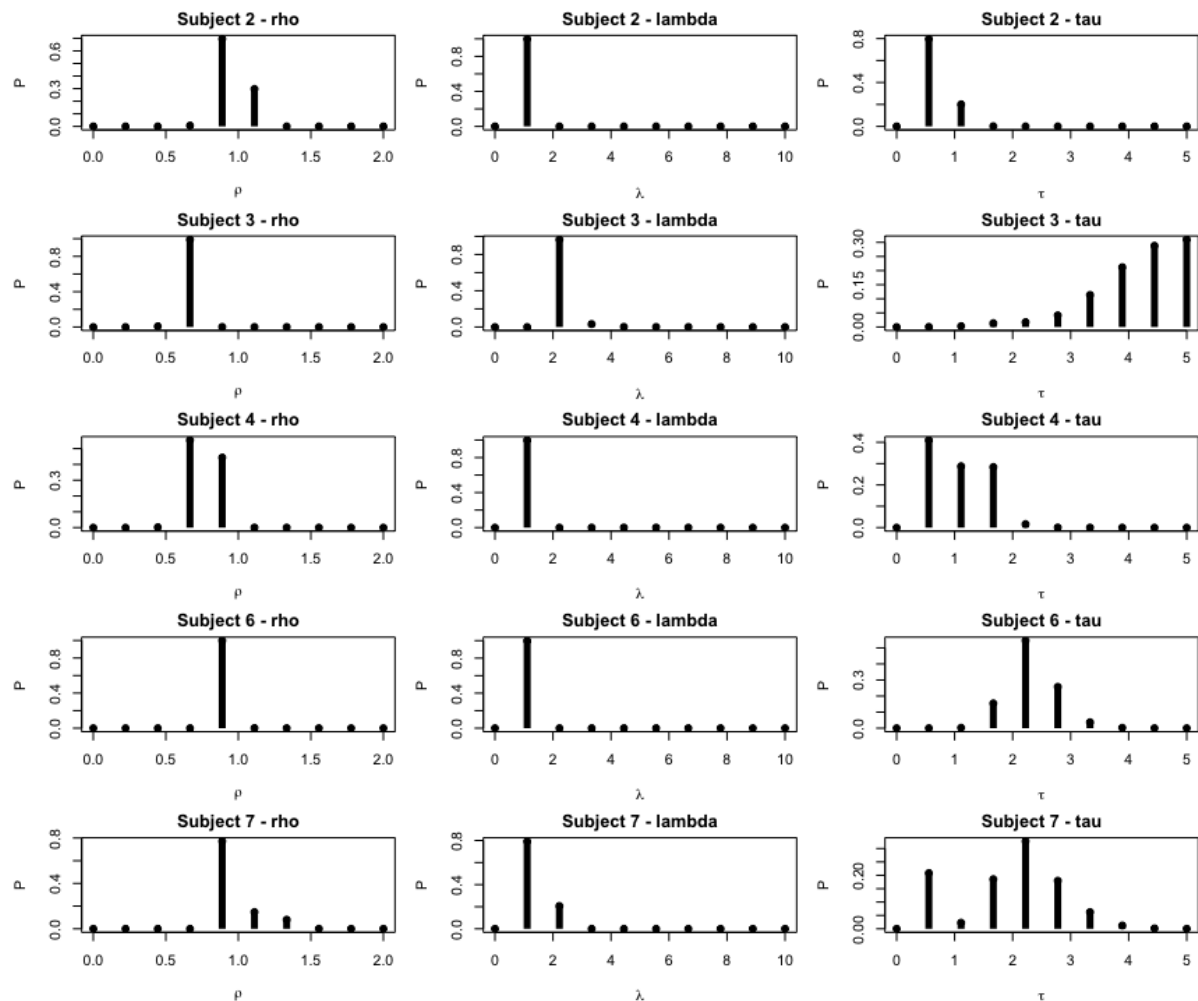
Q4 (4)

Compared to the estimated parameters found in (2) the following scatter plot above from (3) shows an improvement in parameter recovery. Although it is not a significant difference, we can see greater alignment with the subject clusters and the dashed identity line in α_{pos} and α_{neg} recovery. This may have occurred naturally because a greater number of

trials (100->300) leads to more retrieval of information from the data. Furthermore, because the probability of selecting option 1 or 2 has changed from 0.5/0.8 to 0.3/0.6, the options are less extreme and more negative PE trials will be generated from the participants, leading to more data that can be used to estimate α_{neg} . These reasons suggest why better recovery occurred in the latter simulation setting.

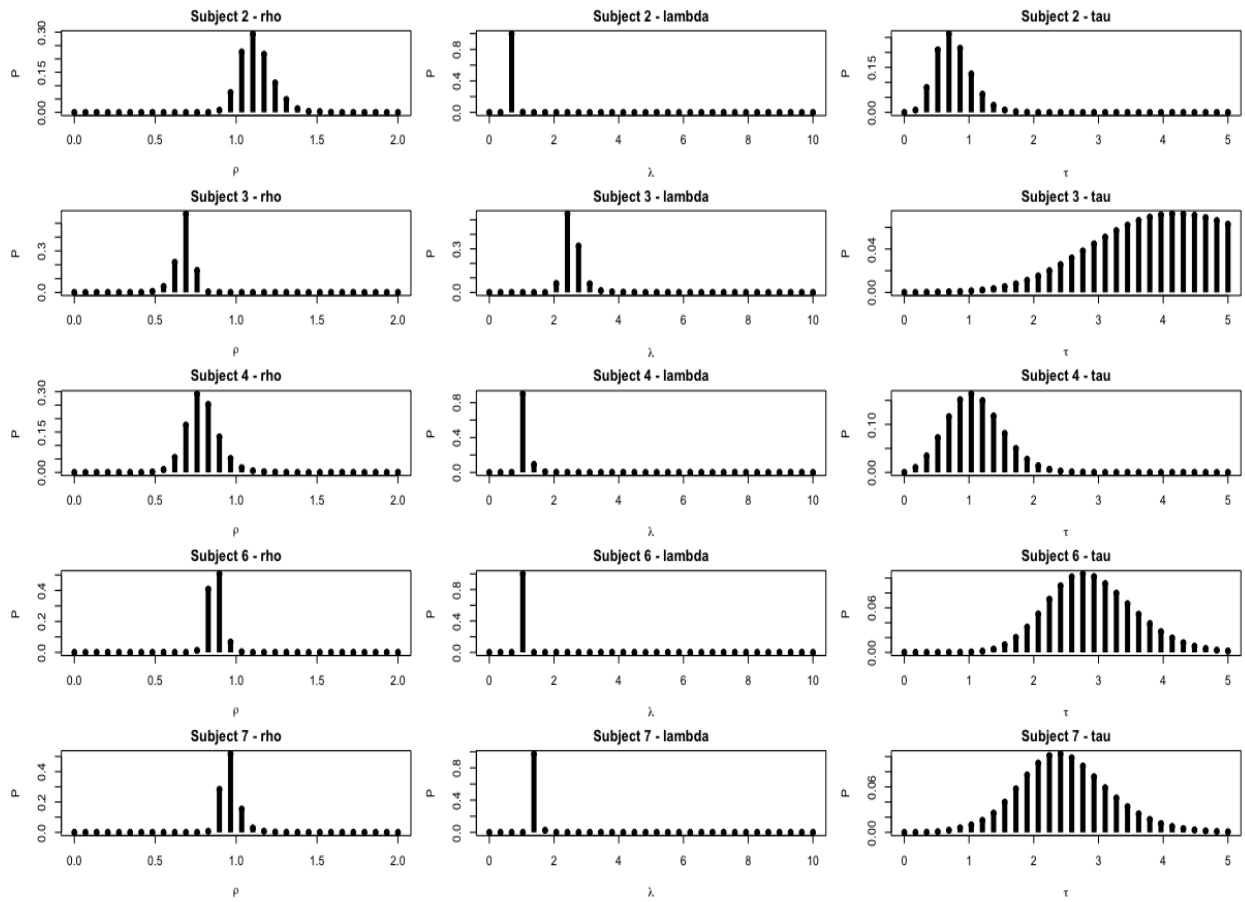
Bonus question: Q5 (1)

Posterior distributions of the ra_prospect model for subjects 2~7



These histograms each show the posterior distribution of risk preference, loss aversion and choice consistency of the ra_prospect model for each participant.

Q5 (2)



This is the posterior distribution using 30 grids per parameter. We can see that compared to (1), the posteriors look smoother due to an increased number of grids. The location of the peak becomes more precise as well due to the posterior modes.